



NAVAL
POSTGRADUATE
SCHOOL

Artificial Intelligence – Project

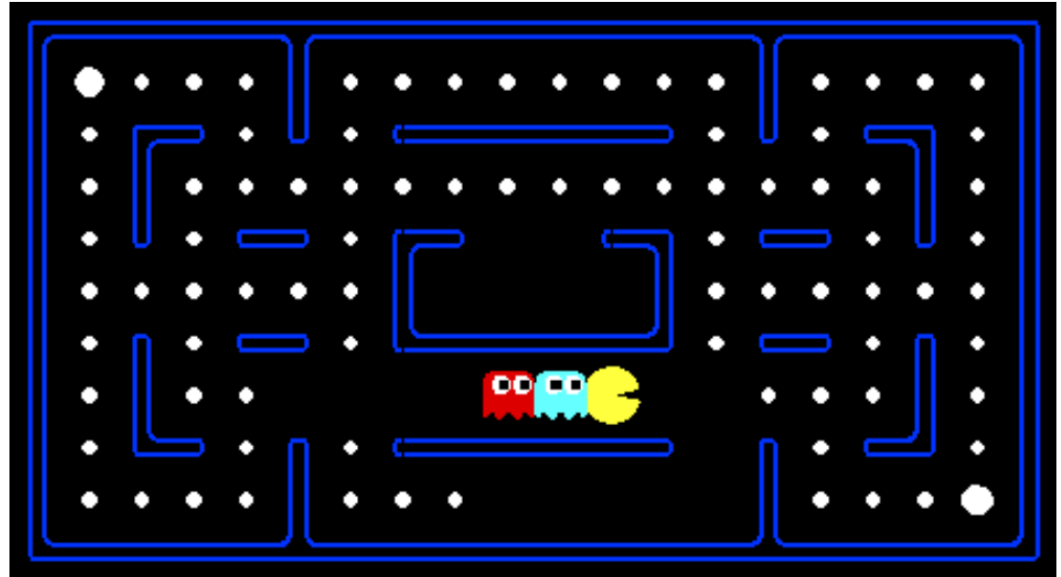
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Pac-Man

- A* Search
- Reflex Agent
- Minimax
- Alpha-Beta Pruning
- Expectimax
- Evaluation Function



- Graded on overall points using python autograder.py
- Competition: Graded on Pac-Man points scored during demo



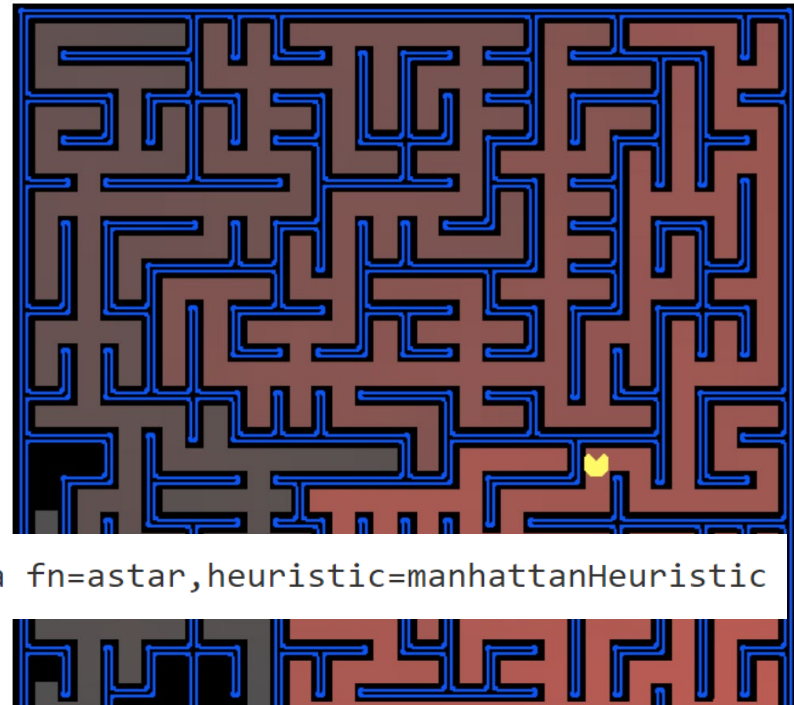
A* Search -3pts

- Download project files
 - <https://inst.eecs.berkeley.edu/~cs188/sp26/projects/proj1/>
- Follow Instructions
- Complete Question 4.
- Ignore all other questions
- Implement your own heuristic
- `run python autograder.py`
 - See points earned
- Run command below to test with MH

```
python pacman.py -l bigMaze -z .5 -p SearchAgent -a fn=astar,heuristic=manhattanHeuristic
```

- Competition: Run command below to test

```
python pacman.py -l bigMaze -z .5 -p SearchAgent -a fn=astar,heuristic= your_heuristic
```





All other algorithms

- Download project files
 - <https://inst.eecs.berkeley.edu/~cs188/sp26/projects/proj2/>
 - Follow the instructions
 - Q1: Reflex Agent
 - Q2: Minimax
 - Q3: Alpha-Beta Pruning
 - Q4: Expectimax
 - Q5: Evaluation Function
-



Python Autograder

- Python autograder.py
- A* Search – 3 pts
- Reflex Agent – 4 points
- Minimax – 5 points
- Alpha-Beta Pruning – 5 points
- Expectimax – 5 points
- Evaluation Function – 6 points



Pac-Man Competition

- A* Search

```
python pacman.py -l bigMaze -z .5 -p SearchAgent -a fn=astar,heuristic= your_heuristic
```

- Reflex Agent

```
python pacman.py -p ReflexAgent -l testClassic
```

- Minimax

```
python autograder.py -q q2
```

- Alpha-Beta Pruning

```
python autograder.py -q q3
```

- Expectimax

```
python autograder.py -q q4
```

- Evaluation Function

```
python autograder.py -q q5
```



Deliverables

- Turn in your code (search.py, multiagents.py)
- Class presentation and Demo
 - ~10 min presentation
 - Your implementations
 - Design decisions
 - Heuristic functions
 - Search depths
- Demo each implementation



Grading

- 90% autograder score
- 10% Presentation

Competition Prizes

1st place: Add 10 points to lowest exam score

2nd place: Add 5 points to lowest exam score

3rd place: Add 3 points to lowest exam score



Teams

- 1-3 people per team
- Poll to gather Team Names and Members